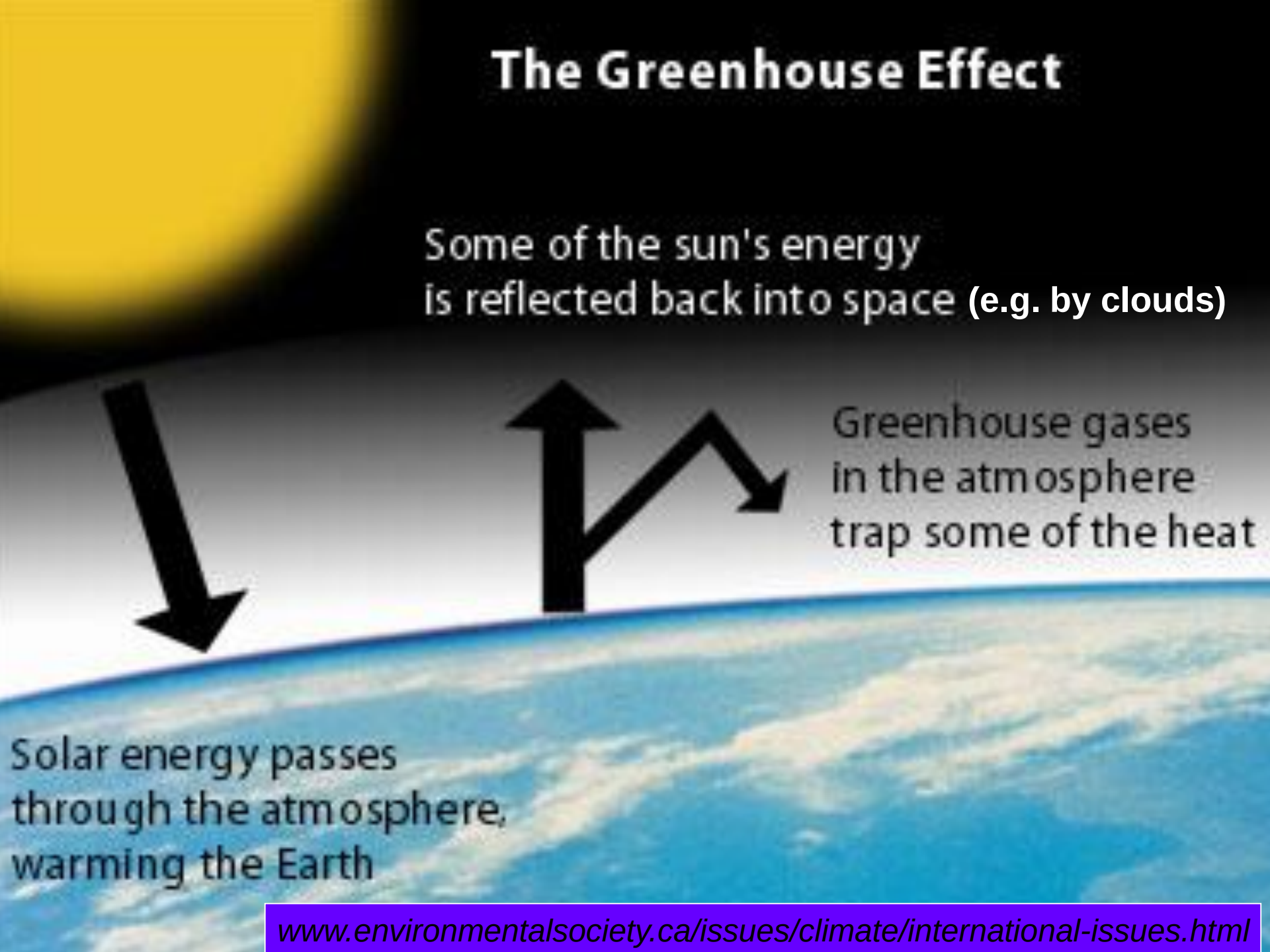
EART11200: Natural Scientist's Toolkit

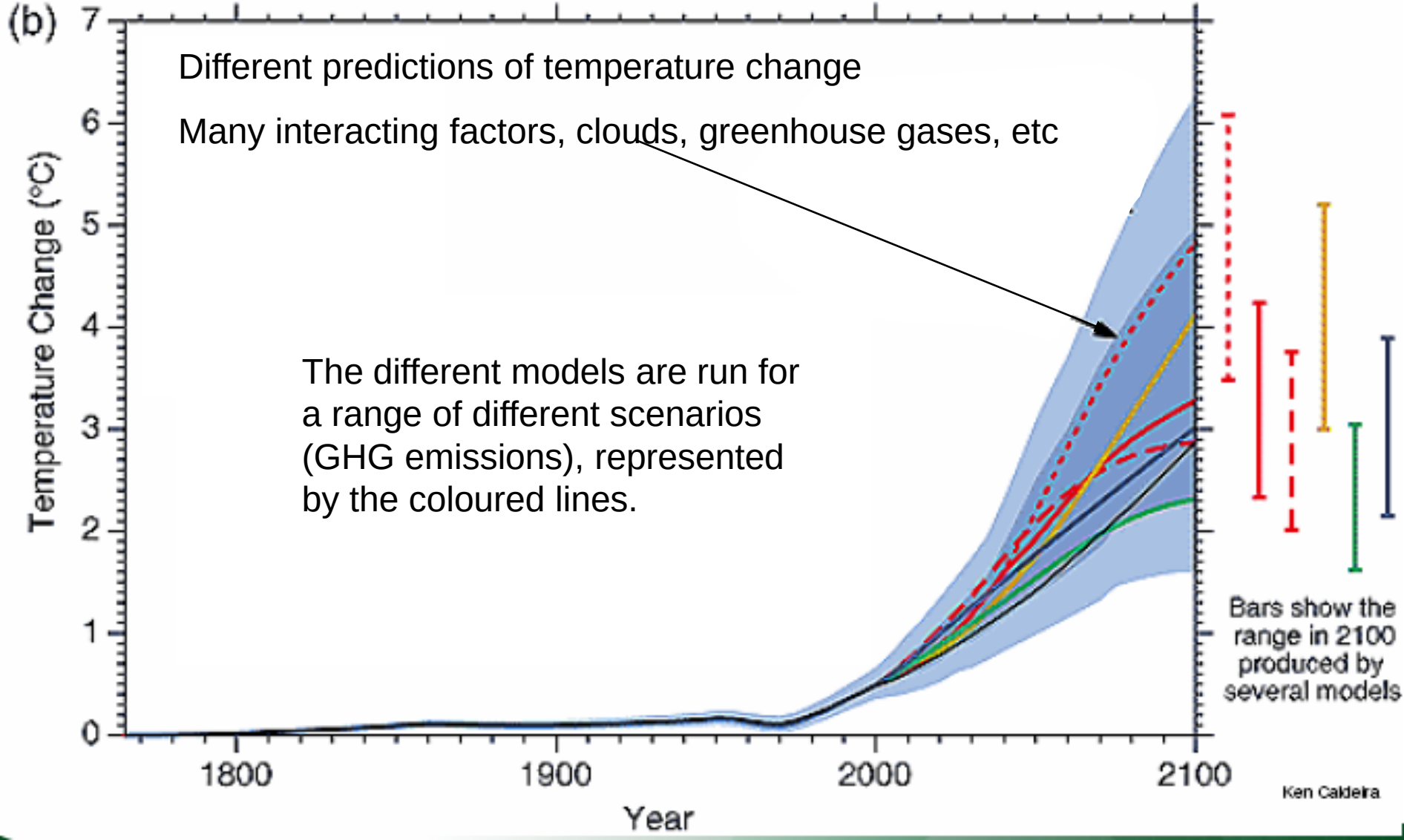


This week we will look at the burning of fossil fuels, and its consequences. The greenhouse effect; the problem.

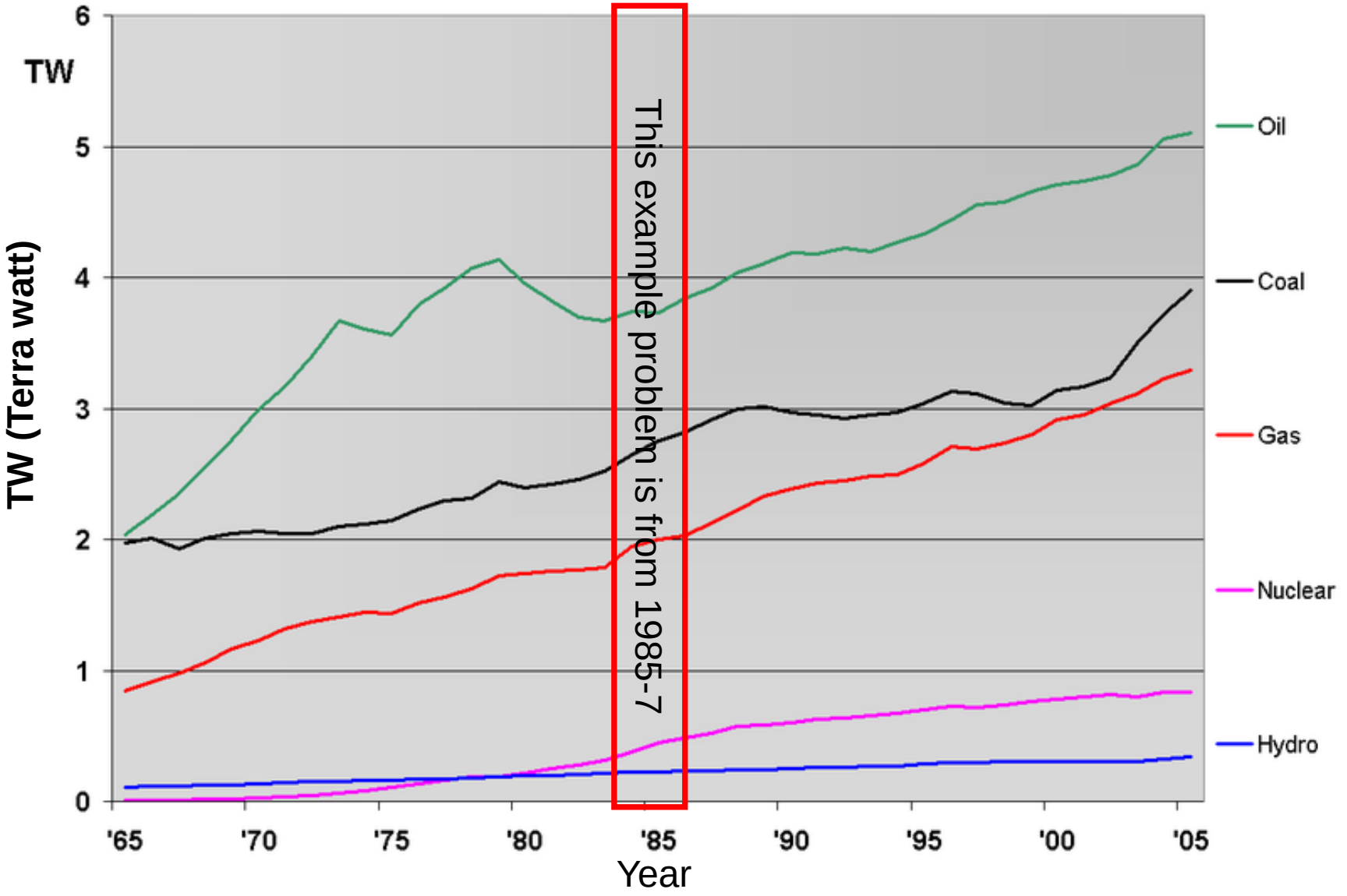


www.environmentalsociety.ca/issues/climate/international-issues.html

Climate predictions show temperatures to increase throughout this century



We can work out how much CO2 was made from fossil fuels by considering the global power consumption



How much CO_2 is made by burning these fuels?

Problem solving 6

Chemical reactions and equilibria part 1

6.1 Altering the Atmosphere by burning fossil fuels

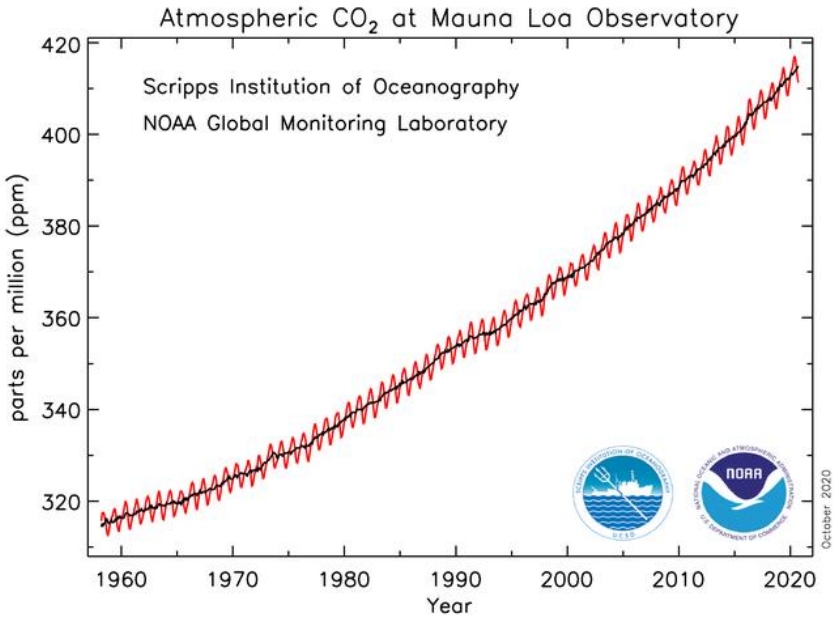
Problem 6.1 *In a recent year how much O_2 is removed from the atmosphere due to the combustion of fossil fuel on Earth, and how much CO_2 and H_2O is produced in the combustion process?*

The primary elemental constituents of the three major types of fossil fuel (natural gas, petroleum, and coal) are carbon and hydrogen. When fossil fuel is burned, oxygen from the atmosphere combines with the carbon to make CO_2 and with the hydrogen to make H_2O. In addition, coal contains some water (typically 10-15% by weight), which is released to the atmosphere upon combustion of the coal. All three types of fossil fuel contain various other substances such as ash, sulfur, and trace metals in even lower concentrations.

We will need to know the chemical composition of the fuel to calculate CO_2 and H_2O emission and the oxygen consumption, we need to balance the chemical combustion reactions for each fuel. Consider petroleum, first, with the approximate chemical composition $CH_{1.5}$. The combustion reaction for this fuel is

$$CH_{1.5} + xO_2 \rightarrow yCO_2 + zH_2O \qquad (6.1)$$

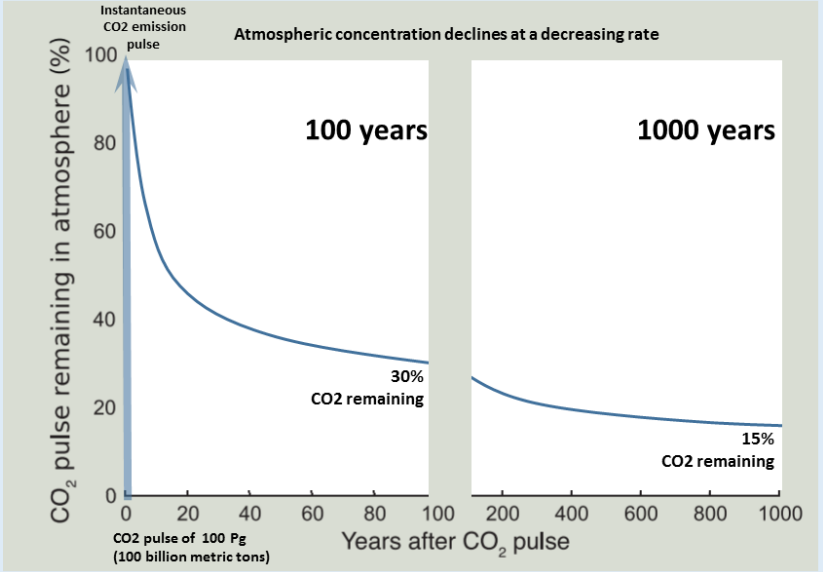
- How does this affect the atmospheric CO_2 concentration?



Why do we not care about water vapour released into the atmosphere from combustion?

Water vapour is also a greenhouse gas…..

CO2's ultra long atmospheric residence time



IPCC 2014 AR5 WG1 TFE.7, Figure 1

Peter Carter Climate Emergency Institute

This week in the practical

- Do supplementary material (additional problems on ppmv / mixing ratio)
- Do main problems: and 1 & 2 (don’t need to do 3&4)
- Homework: Blackboard exercises
  - Altering the Atmosphere
  - Factorising / simplifying fractions